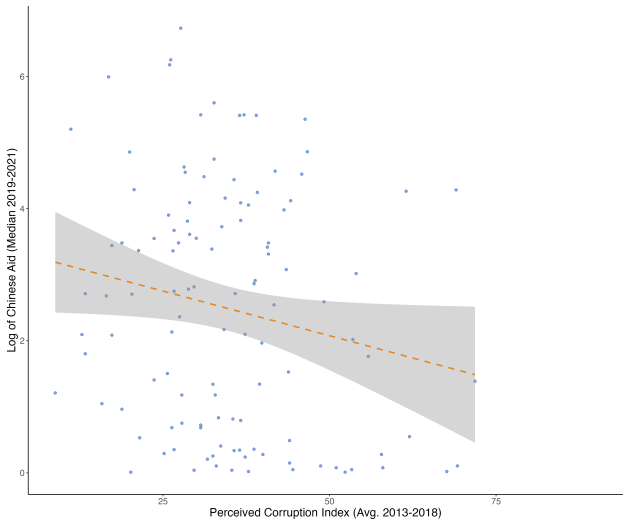
To: Dilma Rousseff, President of the BRICS Bank
From: Lucas Iantorno Klotz, School of Global Policy and Strategy (UC San Diego)
Re: Are corrupt nations more likely to receive Chinese Aid?
Date: March 13, 2026

Introduction

Foreign aid has helped nations narrow the gap with developed countries. The United States has long been a major donor, and such aid is typically conditional on institutional reforms and transparency. Chinese aid, on the other hand, is said to be less transparent, raising an important research question: Are corrupt countries more likely to receive Chinese aid? Figure 1 displays the relationship between a corruption index and the amount of Chinese Aid. A regression line shows a negative slope, indicating that as countries become less corrupt, they receive, on average, less aid. In this memo, we directly address whether **corruption in recipient countries increases the volume of Chinese aid received**. This question matters for understanding the implications of Chinese foreign assistance for global governance standards.

Figure 1: Corrupted countries tend to receive more Chinese Aid, on average



Notes: This figure shows the relationship between the Perceived Corruption Index and the Log of Chinese Aid. Higher index numbers mean less corruption. The dashed orange line represents the regression line. Gray shadow is the 95% CI. N = 116.

Theory and Hypothesis

Developing countries struggle with corruption and request foreign aid to address infrastructure, health, education, and other challenges. Western aid is provided conditionally, requiring transparency and institutional reforms that often undermine local interest groups or expose weak state capacity. As corruption increases, so does elite influence and state weakness. As a result, corrupt countries may rely on more flexible aid from China. Why China offers more flexible terms is beyond the scope of this memo, though political alliances and access to

natural resources (common in developing countries) are plausible explanations. Under this framework, we test the following hypothesis: as a country's corruption levels rise, it will turn to China more often, resulting in higher volumes of development finance.

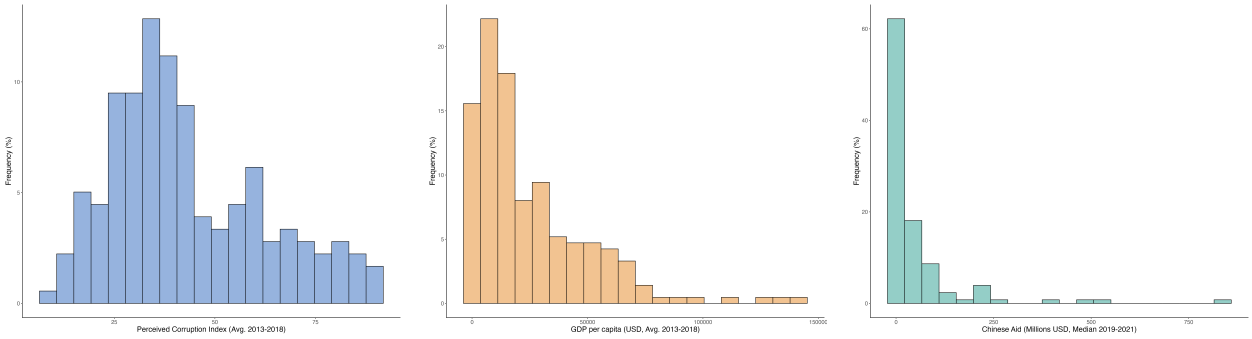
Data Description

We draw on two data sources. The first contains Chinese aid projects, including the amount in inflation-adjusted dollars, country, sector, and aid type. The second, from Our World in Data, contains country-level characteristics: the perceived corruption index, the Gini index, GDP per capita, the share of the population in extreme poverty, and a democracy index. Our variable of interest is the first one[1]. Figure 2a shows the distribution of the corruption index. It has a nearly bimodal distribution, clearly showing two main groups: those who are more corrupt (on the left) and those who are more 'clean' (on the right). This is crucial to find a significant association between the variable and Chinese aid.

We conduct a cross-country analysis, using the 2013–2018 period as a baseline to predict Chinese aid volume over 2019–2021. We exclude suspended, cancelled, and uncommitted projects, as no funds were disbursed in these cases (about 7% of total cases). We also exclude COVID-19-related projects (fewer than 9% of observations), as the pandemic may have altered the flexibility of both Chinese and Western aid, potentially confounding our findings.

Aid volume is aggregated by country median, a conservative measure robust to outliers, for both the baseline and outcome periods. Country characteristics from Our World in Data are aggregated using averages because they exhibit little within-country variation during the baseline period. The median Chinese aid for both periods and GDP per capita were log-transformed to address skewness (see Figures 2b and 2c). Our final dataset includes 116 countries.

Figure 2: Distribution of variables of interest



(a) Corruption Index (b) GDP per capita (2013 - 2018) (c) Chinese Aid

Notes: Distribution of corruption index (N = 179), GDP per capita (N = 212) and Chinese aid (2019 - 2021, N = 127). These histograms consider all countries that have the selected information. Corruption Index and GDP per capita are averages from the baseline period. Chinese aid is the median of the period of interest.

[1]Higher index scores mean less corruption.

Methodology
We estimate the following linear regression:

$$\text{chinese_aid}_i = \beta_0 + \beta_1 \text{corruption_index}_i + \boldsymbol{\delta}' \mathbf{controls}_i + \epsilon_i \tag{1}$$

where chinese_aid_i is the log of the median Chinese aid for 2019–2021 for country i, and $\text{corruption_index}_i$ is the average corruption index for 2013–2018. $\mathbf{controls}_i$ is a set of baseline characteristics selected to address omitted variable bias, and ϵ_i is the error term.

Controls are added sequentially: i) log GDP per capita, as economic development affects both corruption and aid volume; ii) median Chinese aid in the baseline, as current aid is strongly correlated with past aid; iii) share of the population in extreme poverty, which correlates with corruption and is a primary driver of foreign aid demand; iv) the Gini index, as inequality concentrates elite power and signals greater need for aid; and v) a democracy index, as less democratic countries tend to be more corrupt and may be more aligned with China geopolitically.

Several controls capture overlapping dimensions of development, raising concerns about multicollinearity. Breusch-Pagan tests were conducted for each specification, and we found little to moderate heteroskedasticity. Robust standard errors (HC1) are used throughout.

Findings
Column (1) of Table 1 shows that a 1-point increase in the corruption index is associated with a 2.76% average decrease in Chinese aid, significant at the 5% level. A more meaningful comparison is a 10-point increase, associated with an average 24.4% decrease in aid. Given that the average median aid in the sample is approximately 50 million USD, this corresponds to roughly 12 million dollars — a substantively large effect.

In Columns (4) and (5), the negative association remains robust after adding log GDP per capita, log baseline Chinese aid, the share of the population in extreme poverty, and the Gini Index with significance holding at the 10% level. In Column (6), the result loses significance in the fully specified model, which we attribute to two factors: the reduction in sample size to 80 countries and multicollinearity among economic controls. GDP per capita is correlated with corruption (r = 0.72), poverty (r = −0.75), and the democracy index (r = 0.56), making it difficult for the model to isolate each variable's independent association with Chinese aid. Moreover, the model's explanatory power reaches its highest level in Model 3, where the adjusted R2 indicates that 41.7% of the outcome's variation is explained by the variations of the independent variables. Still, the share of people in extreme poverty is a very important control, so we believe Column (4) provides the best specification (more details in the Appendix).

Limitations and Conclusion
The main limitation is sample size. Running the fully specified model required dropping nearly one-third of countries. This reduction in sample size limits statistical power, potentially masking a significant relationship that a larger sample would detect. Moreover, the 36

Table 1: Regression Results

	Log of Median Chinese Aid (2019–2021)					
	(1)	(2)	(3)	(4)	(5)	(6)
Corruption Index	−0.027**	−0.058***	−0.039***	−0.028*	−0.028*	−0.015
	(0.013)	(0.014)	(0.011)	(0.015)	(0.015)	(0.019)
Log GDP per Capita		0.820***	0.477**	0.312	0.328	0.416
		(0.164)	(0.190)	(0.356)	(0.367)	(0.373)
Log Aid Baseline			0.645***	0.683***	0.682***	0.625***
			(0.103)	(0.133)	(0.134)	(0.142)
Extreme Poverty (%)				−0.009	−0.008	−0.009
				(0.015)	(0.015)	(0.016)
Gini Index					−0.360	−0.458
					(2.113)	(2.286)
Democracy Index						−0.104
						(0.136)
(Intercept)	3.427***	−2.901**	−2.102	−1.035	−1.061	−1.475
	(0.468)	(1.329)	(1.322)	(2.902)	(2.907)	(2.944)
Num. Obs.	116	112	109	84	84	80
R^2	0.035	0.179	0.434	0.423	0.423	0.423
Adj. R^2	0.027	0.164	0.417	0.394	0.386	0.376
RMSE	1.81	1.67	1.40	1.41	1.41	1.39

Notes: Regression results estimating the association between corruption and Chinese Aid. The log of average GDP and the log of Median Aid Baseline are both computed from the baseline period (2013-2018). The variable of interest and the controls are averages from the baseline period. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

excluded countries were, on average, less democratic and received less baseline aid, which limits the external validity of our findings. Therefore, results may not generalize to the full population of Chinese aid recipients.

Despite these limitations, we find consistent evidence of a negative association between the corruption index and Chinese aid volume, robust to the addition of key controls. Our findings are not causal; addressing causality would require more complex research designs and data on aid flows from other donors to verify whether systematic differences exist between Chinese and Western funding patterns. Beyond expanding i) sample size and ii) new information on donors, an indication of whether a country is a commodity exporter (e.g., critical minerals or oil) could help test mechanisms that drive Chinese Aid.

Based on these findings, we recommend that the BRICS Bank improve its monitoring of Chinese aid to developing countries. If China is not verifying whether aid is achieving its stated goals, the Bank can perform this oversight function when resources flow through it. Without accountability, Chinese aid risks perpetuating the conditions it aims to address.

Appendix: Regression Diagnostics
In this Appendix, we provide the diagnoses of the regression exercise. We discuss sampling and provide evidence of how the analysis respects the Gauss-Markov Assumptions.

Given concerns about collinearity, sample size, and model explanatory power, we believe that Model 4's specification and results are the primary ones to be considered in the main text.

Sampling
As discussed in the main text, almost a third of the countries were dropped as we increased the number of controls in the regression. This can cause problems like selection bias. Therefore, we performed balance tests on the observable variables. In particular, we test the difference in means (t-tests) between the group of countries with complete information and the group with missing data. For the latter, each variable has a distinct number of countries.

Figure 3 shows that there are no systematic differences across corruption, inequality, extreme poverty, and GDP per capita. This is good, and helps our case that the main limitation of our exercise is only the sample size. There are, however, significant differences in democracy and the baseline Chinese Aid. In particular, the dropped countries are, on average, less democratic and received less aid from China between 2013 and 2018. This hurts the external validity of our findings. We cannot generalize our findings to the whole population of countries that have received Chinese Aid.

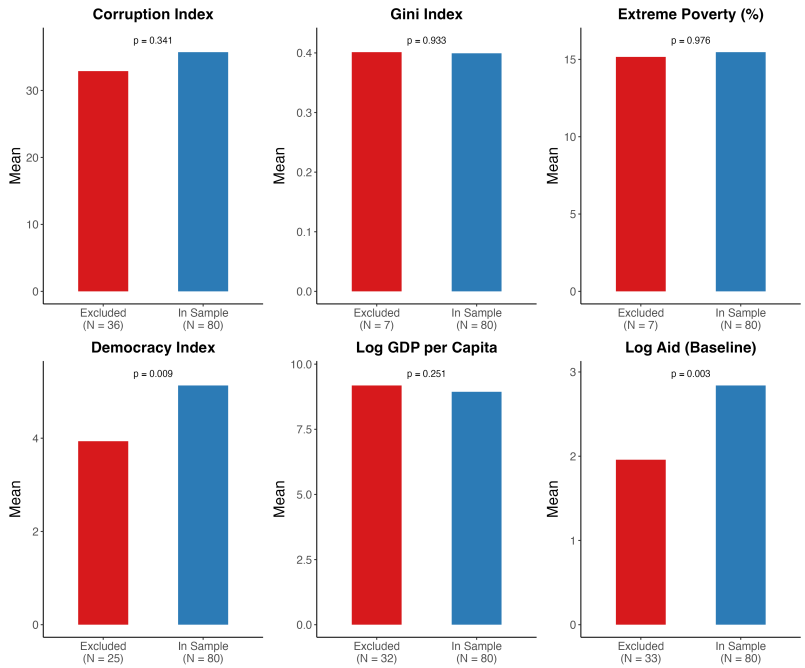
Linearity in the parameters
In this section, we provide evidence that the parameters are linear in relation to outcome y. Figure 4a shows that as we include more controls, residuals get roughly centered around zero (with no systematic patterns) and with the smooth line (blue) close to the dashed line (red). There are no striking changes after model 4.

Figure 4b shows the clear linear relationship of the regressors in relation to the outcome variable. This suggests that there is no need for quadratic forms to estimate the relationship of interest (corruption and Chinese aid).

Normality of the error terms
Figure 5 provides evidence on the normality of the error terms. From the baseline model, one can see that the residuals might be heavy-tailed, but this pattern disappears as we include more controls in the model specification. From Model 3 onward, there's a clear match between the residuals' quantiles and the theoretical quantiles of a normal distribution.

Checking for Omitted Variable Bias
Figure 6 displays the residuals of the variable of interest, corruption index, across the six model specifications. One can see that the residuals are overall centered around zero with no clear patterns. Randomly spread errors provide supportive evidence that we are not omitting a variable from the model that might bias our estimate.

Figure 3: Balance tests: dropped countries due to missing values



Notes: This figure shows the average of the variable of interest and the controls, by Excluded (red) and In Sample (blue) groups. *In Sample* refers to the 80 countries with complete data across all model variables. *Excluded* refers to countries missing at least one control variable but present in the baseline sample (N = 116). P-values correspond to two-sided t-tests of equality of means between groups.

Checking for Heteroskedasticity and Multicolinearity

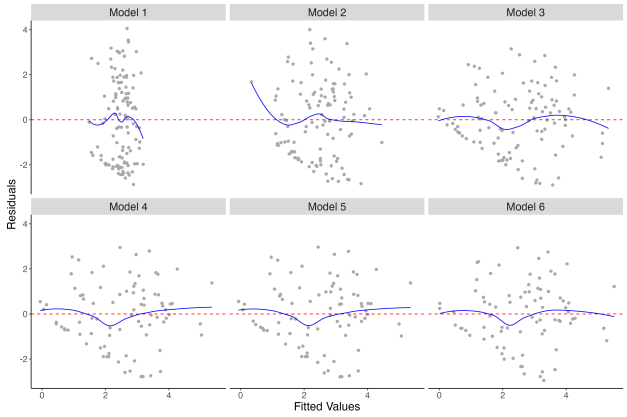
To detect heteroskedasticity, or non-constant variance of residuals, we applied the Breusch-Pagan test to all model specifications. For all of them, it failed to reject the null hypothesis that the error's variance is constant across the fitted values. Still, our regression tables display robust standard errors to account for possible heteroskedasticity.

To detect multicollinearity, we calculated the Variance Inflation Factor (VIF), which measures how much of a coefficient's variance increases due to colinearity. For models 1 to 3, the coefficients' VIFs are around 1, suggesting no multicollinearity. Model 4 shows slightly greater collinearity, driven mainly by GDP per capita and the share of extreme poverty (3.88 and 2.77, respectively). This is still low, but the increase is expected, as the two tend to be highly correlated. In model 5, GDP per capita shows moderate collinearity, as its VIF increases to approximately. 4.34 and the share of extreme poverty to app. 3.7. Model 6 presents similar results to the latter. The latest variable, the democracy index, has a VIF of 2.44.

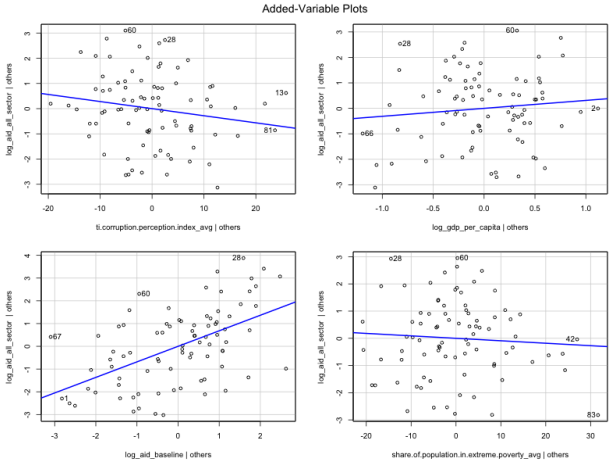
Considering that GDP per capita is often correlated with variables such as corruption, poverty, inequality, and political regime, the increase in collinearity as we reach the full model and sample size suggests that the model has difficulty isolating the effect of each variable on Chinese Aid. For that reason, we chose to focus on model 4, which has little colinearity and includes the main controls.

Figure 4: Validating the linearity of the parameters

(a) Fitted values vs. Residuals, by model specification

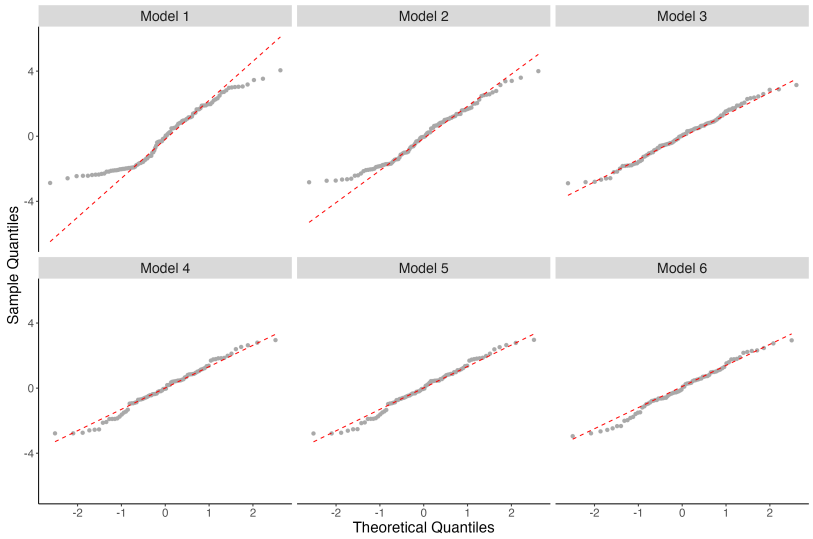


(b) Marginal effect of relevant variables, controlling for the remaining variables



Notes: Figure 3a shows the relationship between each model's fitted values (x-axis) and their corresponding residuals (y-axis). The red dashed line represents the zero line. The blue line helps to see the patterns of the distribution. Figure 3b shows the marginal effects of the explanatory variables in model 4, conditional on the remaining variables. The blue line shows the average linear relationship, as expected.

Figure 5: Validating the Normality of Errors Assumption



Notes: This figure shows a quantile-quantile plot (QQ) to check the normality of the errors' distribution. Each plot follows a model specification from Table 1. Red diagonal line shows the match between the residuals' quantile (y-axis) and theoretical quantiles of a normal distribution (x-axis).

Removing Outliers

Following our decision in the previous section, we checked how many observations were identified as outliers under model 4's specification. We defined outliers using four rules:

- Studentized Residuals: 3 observations.
- Leverage: 7 observations.
- Cook's Distance: 4 observations.
- Difference in Fitted Values (DFFITS): 4 observations.

Four observations were flagged in multiple rules: Ethiopia, Guinea, Serbia, and Zambia.

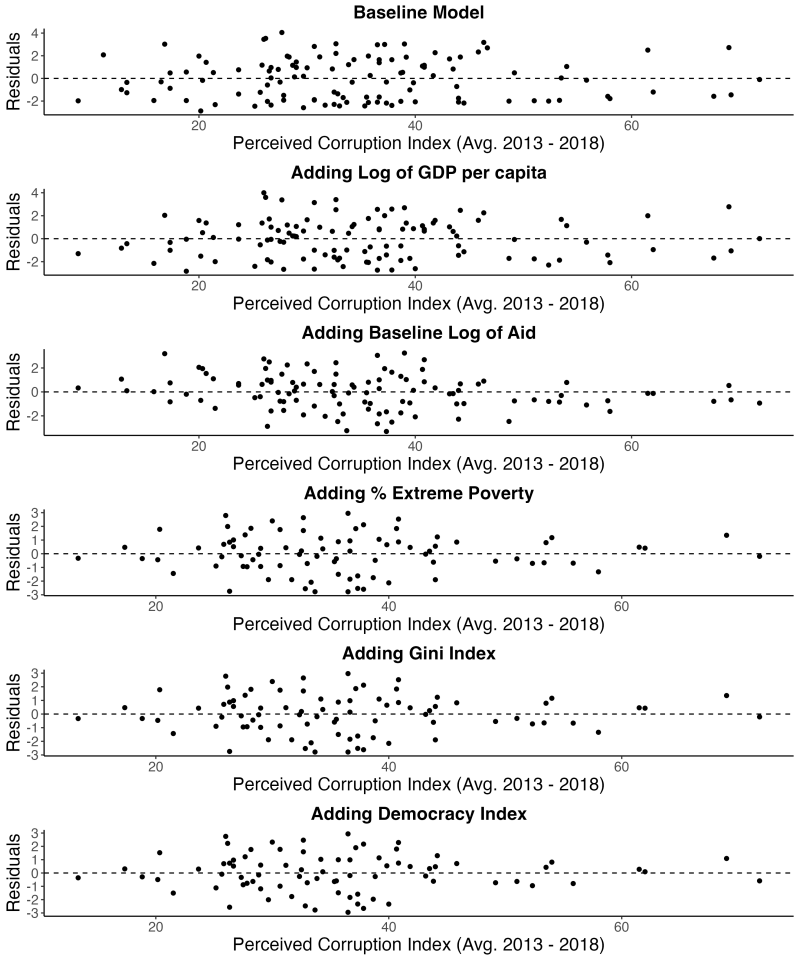
Table 2 shows model 4's specifications with (column 1) and without (column 2) outliers. Any country flagged in any outlier measure was dropped (11 in total). Our variable of interest, the corruption index, loses the 10%-level statistical significance when we drop these observations. As a placebo test, we randomly removed 11 countries, and again, the coefficient is not statistically significant (column 3). Therefore, we cannot imply that the loss of statistical significance is due to the outliers. It might be a problem of statistical power due to the sample size.

Table 2: Regression Results (removing outliers + placebo test)

	Log of Median Chinese Aid (2019–2021)		
	(1)	(2)	(3)
Corruption Index	−0.028*	−0.021	−0.019
	(0.015)	(0.017)	(0.014)
Log GDP per Capita	0.312	0.270	0.101
	(0.356)	(0.326)	(0.358)
Log Aid Baseline	0.683***	0.836***	0.801***
	(0.133)	(0.122)	(0.130)
Extreme Poverty (%)	−0.009	0.013	−0.015
	(0.015)	(0.017)	(0.016)
(Intercept)	−1.035	−1.651	0.371
	(2.902)	(2.577)	(2.953)
Num. Obs.	84	73	73
R^2	0.423	0.503	0.476
Adj. R^2	0.394	0.473	0.445
RMSE	1.41	1.28	1.37

Notes: Regression results estimating the association between corruption and Chinese Aid. Model 2 has fewer observations as they were removed based on four distinct measures of identifying outliers. Model 3 is a placebo test, randomly removing the same number of countries as in Model 2. The log of average GDP and the log of Median Aid Baseline are both computed from the baseline period (2013-2018). The variable of interest and the controls are averages from the baseline period. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 6: Residual plots: evidence of zero mean conditional assumption



Notes: This figure shows the residual plots from all models' specifications (see Table 1). Each analysis adds one control variable. Due to data limitations, each analysis has a distinct sample size.

Acknowledgments

For this work, in addition to previous exposure to R programming as both a graduate Economics student in Brazil and a research assistant at Stanford University, and during QM II lectures, labs, and TA office hours, I, Lucas I. Klotz, acknowledge that I used Claude AI to i) discuss my findings, ii) improve my theoretical reasoning and coding. In particular, Claude was very useful in helping out with functions that increase the efficiency of my code and save time. Claude also helped me reduce my word count from 1,600 to 1,100. Grammarly was used to correct and improve the writing. All interpretations, analyses, and conclusions in this assignment are my own, and I completed this work independently.